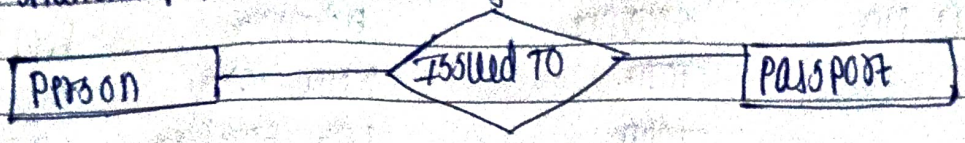
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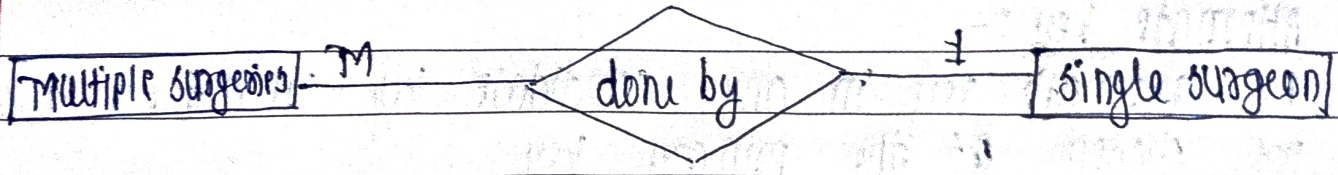
★ Cardinality In ER Model :-

1) One-to-one :-
- When each entity in each entity set can take part only once in the relationship, the cardinality is one to one.

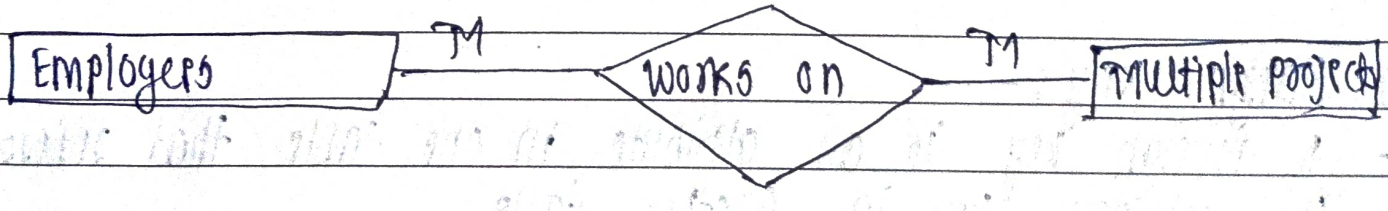


2) One-to-Many :-
- In one-to-many relationship, one entity can be associated with multiple entities.
- Ex:- Single surgeon department can have many doctors.

3) Many-to-one :-
- When entities in one entity set can take part only once in the relationship set and entities in other entity sets can take part more than once in relationship set, cardinality is many to one.
- Ex:-



4) Many to Many :-
- When entities in all entity sets can take part more than once in the relationship cardinality is many to many.



★ Key In Relational Model :-

- Key are fundamental elements of the relational database model that ensures uniqueness, data integrity, and efficient access.
- They uniquely identify each row in a table.

1) Candidate key :-
- The minimal set of attributes that can uniquely identify a tuple is known as candidate key.
- EX :-

STUD_NO	SName	Address	Phone
1	TOM	New York	1234567890
2	John	Los Angeles	2233445566
3	Alice	Chicago	1745886655

For above table STUD_NO is candidate key as it uniquely identifies each record.

2) Super key :-
- The set of one or more attributes that can uniquely identify a tuple is known as ~~can~~ super key.
- EX :- A super key could be combination of STUD_NO & Phone as this combination uniquely identifies a student.

3) Alternate key :-
- An alternate key in any candidate key in a table that is not chosen as the primary key.
- EX :- In above table STUD_NO, & PHONE are candidate key. If STUD_NO is choosen as primary key, then phone can be considered as an alternate key.

4) Foreign key :-
- A foreign key is an attribute in one table that refers to the primary key in another table.

5) Partial key :-
- A partial key is chosen from a weak entity to help in records, but it can't uniquely identify a record by itself.

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6) Primary key :-
- A primary key is chosen from the set of candidate keys to uniquely identify each record in table.

7) Secondary keys :-
- A secondary key is an attribute or a combination of attributes used to search or query records in a table, but it doesn't guarantee uniqueness.

8) Unique key :-
- A unique key is a database constraint that ensures that all values in a specific column or a combination of columns are unique across all the rows in a table.

9) Composite key :-
- Two or more attributes are used together to make a composite key

10) Surrogate key :-
- A surrogate key is an artificial attribute created to uniquely identify each record in a table when no suitable natural key is available.